MITRC

DEPARTMENT OF CIVIL ENGINEERING

Name of Faculty: Karishma Mittal
Name of Subject: FHI
Semester: VIII
Branch: CE, EE, ME CSE, AIDS, AIML
Unit No: 6

Subject Code: 8AR6.601
Session: 2025-26 (Even Sem)
Batch: A&B
Date of submission: 18/2/26

Lecture No.	Topic Covered	Page No.	Total Page
1	Modes of Communication, Communication Within Teams	2-4	3
2	Communication Between Teams, Challenges in Inter-Team Communication & Solutions, Difference Between Communication Within Teams and Communication Between Teams Communication Problems	5-8	4
3	How to Overcome Communication Problems, Work Logging and Recording, Manual, Handbook	9-11	3
4	Technical Paper, Keeping Up-to-Date and Currency in Information	12-15	4
5	Purpose of Information Dissemination, Methods of Disseminating Information	16-17	2

References:

- "Industrial Guide to Ergonomics Engineering", Dr. Osama Mohammed Elmardi Suleiman Khayal
- Mechanical Engineering Department, Faculty of Engineering and Technology, Nile Valley University,
- Atbara, Sudan, 2019
- "Human Performance and Limitations", Oxford Aviation Academy (UK) Limited 2008

Faculty HOD Dean of Academics

Lecture-1

Communication

Modes of Communication

Modes of communication refer to the different ways in which information is transmitted between individuals or groups. These can be broadly classified into the following categories:

1. Verbal Communication

- Oral Communication: Face-to-face conversations, phone calls, video calls, speeches, presentations, etc.
- Written Communication: Emails, letters, reports, memos, text messages, etc.

2. Non-Verbal Communication

- Body Language: Gestures, posture, facial expressions, eye contact, etc.
- Paralanguage: Tone, pitch, volume, and speed of speech.
- Proxemics: Use of personal space in communication.

3. Visual Communication

- Images & Symbols: Signs, charts, graphs, infographics, and maps.
- Videos & Animations: Used in advertising, education, and digital media.

4. Digital Communication

- Emails, Messaging Apps, and Social Media: Platforms like WhatsApp, Twitter, LinkedIn, and Slack.
- Web-Based Communication: Websites, blogs, webinars, and online forums.

5. Formal and Informal Communication

- Formal: Official channels like meetings, reports, and business emails.
- Informal: Casual conversations, workplace chats, and social interactions.

Communication Within Teams

Communication within teams refers to the exchange of information, ideas, and feedback among team members to achieve common goals efficiently. Strong team communication fosters collaboration, enhances productivity, and minimizes misunderstandings.

Types of Team Communication

a) Formal Communication

Used for structured and professional interactions.
Examples:

- Team meetings
- Progress reports
- Emails and official memos
- Project management tools

b) Informal Communication

Casual interactions that build relationships and improve teamwork.
Examples:

- Workplace chats (Slack, WhatsApp)
- Coffee break discussions
- Team-building activities

c) One-Way vs. Two-Way Communication

- **One-Way Communication:** Information flows from one party to another without feedback (e.g., announcements, memos).
- **Two-Way Communication:** Allows for discussion and feedback (e.g., brainstorming sessions, Q&A meetings).

d) Vertical vs. Horizontal Communication

- **Vertical Communication:** Between team members at different levels (e.g., manager to employee).
- **Horizontal Communication:** Between team members at the same level (e.g., colleagues working together).

Modes of Communication in Teams

a) Verbal Communication

- **Face-to-Face Conversations** – Quick problem-solving, collaboration.
- **Meetings & Video Calls** – Regular check-ins, brainstorming.

- **Phone Calls** – Urgent discussions, clarifications.

b) Written Communication

- **Emails & Reports** – Detailed updates, official documentation.
- **Instant Messaging** – Quick updates, casual discussions.
- **Project Management Tools** – Task tracking and coordination.

c) Non-Verbal Communication

- **Body Language & Facial Expressions** – Reinforce messages in meetings.
- **Active Listening** – Shows engagement and respect.

d) Digital Communication

- **Collaborative Platforms** – Google Docs, Notion for shared work.
- **Online Discussion Forums** – For ongoing team discussions.

Importance of Effective Team Communication

- **Enhances Collaboration** – Encourages teamwork and problem-solving
- **Prevents Misunderstandings** – Reduces confusion and errors.
- **Improves Productivity** – Streamlines workflows and decision-making.
- **Boosts Employee Morale** – Encourages openness and trust.

Reference

- https://www.slideshare.net/slideshow/human-factors-topic-1-introduction/18379460#11
- https://www.slideshare.net/slideshow/human-factors-7950513/7950513
- **Human Factor Printed by L.N.V.M. Society Group of Institutes**

Lecture-2

Communication Between Teams

Communication between teams refers to the exchange of information, ideas, and resources across different teams within an organization. Effective inter-team communication ensures collaboration, minimizes silos, and aligns teams toward shared goals.

Types of Inter-Team Communication

a) Cross-Functional Communication

Occurs when teams with different expertise work together on a common project.

Example: The marketing team collaborates with the product development team for a new product launch.

b) Sequential Communication

One team's work serves as input for another team.

Example: The software development team completes a feature, and the quality assurance team tests it.

c) Reciprocal Communication

Continuous exchange of feedback and updates between teams.

Example: The customer support team shares client concerns with the product team for improvements.

d) Hierarchical Communication

Occurs when teams communicate across different levels of authority.

Example: The finance team reports budget allocations to project managers.

Challenges in Inter-Team Communication & Solutions

Challenge	Solution
Information Silos	Regular cross-team sync-ups, shared documentation.
Misalignment on Goals	Clearly define objectives and responsibilities.

Communication Overload	Use structured updates instead of excessive messages.
Lack of Clarity	Standardize reporting formats and documentation.
Time Zone & Cultural Differences	Use asynchronous communication and recorded meetings.

Difference Between Communication Within Teams and Communication Between Teams

Aspect	Communication Within Teams	Communication Between Teams
Definition	Exchange of information among members of the same team.	Exchange of information between different teams in an organization.
Purpose	Enhances team collaboration, coordination, and task completion.	Ensures alignment, collaboration, and workflow continuity between teams.
Participants	Team members working on the same project or tasks.	Different teams with distinct roles but interconnected goals.
Communication Flow	Can be informal or formal, mostly horizontal or vertical within the team.	Mostly formal, requiring structured coordination between teams.
Examples	- Daily stand-up meetings - Team discussions on Slack - Task assignments in Trello	- Marketing team working with Sales to plan a campaign - IT team collaborating with HR to implement a new system
Challenges	- Misunderstandings within the team - Lack of clarity in roles - Poor feedback loops	- Information silos between teams - Misalignment on priorities - Dependency delays
Tools Used	Slack, Microsoft Teams, Google Docs, Asana	Emails, Project Management Software (Jira, Trello), Cross-team Meetings
Best Practices	- Encourage open feedback - Define clear team roles - Use collaborative tools effectively	- Regular inter-team syncs - Shared documentation and reporting - Clearly defined responsibilities

Communication Problems

Communication problems refer to obstacles that prevent effective information exchange between individuals, teams, or organizations. These problems can lead to misunderstandings, conflicts, decreased productivity, and poor decision-making

Common Types of Communication Problems

1. Verbal Communication Issues

- **Lack of Clarity** – Messages are leading to misunderstandings.
- **Speaking Too Fast or Too Slow** – Affects message comprehension.
- **Poor Listening Skills** – Ignoring or misinterpreting information.

2. Written Communication Issues

- **Unclear or Ambiguous Writing** – Emails, reports, or instructions that are confusing.
- **Lack of Proper Formatting** – Walls of text with no structure.
- **Grammar & Spelling Errors** – Causes misinterpretation and reduces professionalism.

3. Non-Verbal Communication Issues

- **Negative Body Language** – Crossed arms, lack of eye contact, or defensive posture.
- **Mismatched Verbal & Non-Verbal Cues** – Saying "I'm fine" with an angry tone.
- **Cultural Differences in Gestures** – A thumbs-up may mean approval in one culture but be offensive in another.

4. Technological Communication Issues

- **Technical Glitches** – Poor internet connection in virtual meetings.
- **Over-Reliance on Digital Tools** – Misinterpretation of messages due to lack of tone in emails or texts.
- **Too Many Communication Channels** – Information overload from emails, chat apps, and calls.

5. Organizational Communication Issues

- **Information Silos** – Teams or departments not sharing critical information.
- **Lack of Feedback Mechanisms** – Employees unsure about expectations or performance.
- **Ineffective Meeting Practices** – Too many meetings with no clear outcomes.

Reference

- https://www.slideshare.net/slideshow/human-factors-topic-1-introduction/18379460#11
- https://www.slideshare.net/slideshow/human-factors-7950513/7950513
- **Human Factor Printed by L.N.V.M. Society Group of Institutes**

Lecture-3

How to Overcome Communication Problems?

- **Practice Active Listening** – Focus on the speaker, ask questions, and confirm understanding.
- **Use Clear & Concise Language** – Avoid jargon and be direct.
- **Adapt to Your Audience** – Consider language proficiency, cultural background, and context.
- **Improve Written Communication** – Proofread and format messages properly.
- **Leverage the Right Tools** – Choose appropriate communication channels (e.g., face-to-face for important discussions).
- **Encourage Open Communication** – Create an environment where feedback is welcomed.

Work Logging and Recording

Work logging and recording is the process of systematically documenting work-related activities, tasks, time spent, and progress. It helps track productivity, improve accountability, and streamline project management.

Purpose of Work Logging and Recording

1. **Tracks work progress** – Ensures tasks are completed on time.
2. **Improves accountability** – Helps employees and teams take responsibility for their work.
3. **Enhances productivity** – Identifies bottlenecks and optimizes workflows.
4. **Supports reporting & compliance** – Keeps records for audits, legal requirements, and payroll.
5. **Facilitates collaboration** – Enables transparency between team members and departments.

Types of Work Logging and Recording

a) **Time Logging-**Tracks hours worked on tasks or projects. *Example:* An employee logs daily work hours using Clockify.
b) **Task Logging-**Records completed, ongoing, and pending tasks. *Example:* A developer updates Jira after finishing a coding task.
c) **Incident & Issue Logging-**Documents errors, accidents, or technical issues. *Example:* An IT team logs system failures in a ticketing system.

d) **Meeting & Communication Logging-**Summarizes discussions, decisions, and action items.
Example: A manager records meeting minutes in Google Docs.
e) **Performance Logging-**Monitors employee productivity and goal completion. *Example:* A sales team logs leads and conversions in Salesforce.

Benefits of Work Logging & Recording

- **Efficiency** – Helps teams stay organized and on schedule.
- **Transparency** – Provides clear visibility into workloads and progress.
- **Compliance** – Ensures proper documentation for audits and reporting.
- **Better Decision-Making** – Uses data to improve workflows and resource allocation.

Manual, Handbook, and Technical Paper: Definitions and Differences

Each of these documents serves a unique purpose in providing information, guidance, and technical knowledge. Below is a breakdown of their definitions, purposes, and key differences.

1. Manual

A manual is a document that provides step-by-step instructions on how to operate, use, or maintain a product, system, or process.

Purpose:

- Guides users on proper usage.
- Ensures safety and efficiency.
- Provides troubleshooting steps.

Examples:

- **User Manual:** Instructions for setting up a smartphone.
- **Maintenance Manual:** Guide for servicing machinery.
- **Training Manual:** Procedures for onboarding employees.

Typical Structure:

1. Introduction
2. Step-by-Step Instructions
3. Troubleshooting Guide
4. Safety Precautions

2. Handbook

A handbook is a reference document that provides general guidelines, policies, and best practices on a specific subject.

Purpose:

- Serves as a quick reference for users.
- Provides standardized guidelines and policies.
- Offers comprehensive but easy-to-follow information.

Examples:

- **Employee Handbook:** Company policies and work procedures.
- **Student Handbook:** School rules and academic guidelines.
- **Safety Handbook:** Workplace safety regulations.

Typical Structure:

1. Overview/Introduction
2. Policies & Procedures
3. Guidelines & Best Practices
4. Roles & Responsibilities
5. Resources & References

Reference

- https://www.slideshare.net/slideshow/human-factors-topic-1-introduction/18379460#11
- https://www.slideshare.net/slideshow/human-factors-7950513/7950513
- **Human Factor Printed by L.N.V.M. Society Group of Institutes**

Lecture-4

3. Technical Paper

A technical paper is a detailed, research-based document that presents technical findings, theories, or innovations in a specific field.

Purpose:

- Communicates scientific or engineering research.
- Provides in-depth analysis and data-driven insights.
- Contributes to academic or industry knowledge.

Examples:

- **Engineering Paper:** A study on concrete strength in civil engineering.
- **Scientific Paper:** Research on renewable energy technologies.
- **Medical Paper:** Findings on a new drug's effectiveness.

Typical Structure:

1. Abstract (Summary)
2. Introduction (Problem Statement)
3. Methodology (Research Approach)
4. Results (Findings & Data)
5. Discussion (Analysis & Interpretation)
6. Conclusion (Summary & Future Work)
7. References (Citations & Sources)

Key Differences

Aspect	Manual	Handbook	Technical Paper
Purpose	Step-by-step instructions	Guidelines & policies	Research findings & analysis
Audience	Users & operators	Employees, students, professionals	Researchers, scientists, engineers
Content Type	How-to guides & troubleshooting	Policies, rules & best practices	Data, theories & experimental results
Examples	Device user manual	Employee handbook	Scientific journal article

Keeping Up-to-Date and Currency in Information

Keeping up-to-date refers to staying informed about the latest developments in a particular area or field. This can include industry trends, new technologies, scientific research, regulations, and market shifts.

Key Aspects of Keeping Up-to-Date

- **Continuous Learning:** Regularly acquiring new knowledge through courses, seminars, or self-study.
- **Networking:** Engaging with industry peers, attending conferences, or participating in online forums**.**
- **Accessing Current Resources:** Subscribing to journals, news outlets, and publications to stay informed about recent changes.

Currency refers to how recent or current a piece of information is. In terms of research, reports, or any professional field, the currency of the data is essential because outdated or old information may lead to incorrect conclusions or decisions.

Factors Impacting Currency:

- **Time Sensitivity:** How recent the data is, especially in fast-evolving fields like technology, healthcare, or business**.**
- **Relevance:** Ensuring that information is applicable to current standards and practices.
- **Accuracy:** Ensuring that the information is valid and reflects the most recent findings or advancements.

How to Keep Information Up-to-Date and Current

a) Use Trusted Sources

- **Research Journals & Databases**: Academic journals (like JSTOR, IEEE Xplore) or industry-specific databases provide the latest research and data.
- **News Outlets & Industry Reports**: Subscribing to well-regarded industry publications (e.g., The Wall Street Journal, TechCrunch) helps stay informed about emerging trends.

b) Participate in Professional Development

- **Training Programs & Workshops**: Regularly attending professional development sessions ensures that you are familiar with the latest tools and methodologies in your field.

- **Certifications**: Obtaining or renewing certifications keeps you up-to-date on industry standards and practices.

c) Follow Key Industry Leaders and Influencers

- **Social Media & Blogs**: Following influential figures in your field on platforms like LinkedIn, Twitter, or specialized blogs can help you track breaking news or discussions.
- **Webinars & Podcasts**: Many professionals share insights and trends through webinars or podcasts, offering an easy way to keep updated.

d) Collaborate with Colleagues and Peers

- **Cross-Team or Cross-Industry Collaboration**: Working with others in your field allows you to learn from their experiences and knowledge of recent trends and best practices.
- **Internal Knowledge Sharing**: Regular knowledge-sharing sessions within teams or departments can help ensure that everyone stays up-to-date with the latest changes or updates in the industry.

e) Regularly Update Your Tools and Systems

- **Software Updates**: Ensure that any software tools, project management systems, or databases you use are kept up-to-date to ensure they are utilizing the latest features, security patches, and improvements.
- **Standard Operating Procedures (SOPs)**: Regularly review and revise internal SOPs to reflect new methods, standards, and regulations.

Dissemination of information

Dissemination of information refers to the process of distributing or communicating information to the intended audience. It is a key element in ensuring that knowledge, findings, updates, or instructions reach the right people in a timely and effective manner. Proper dissemination ensures that information is not only shared but also understood and acted upon.

Reference

- https://www.slideshare.net/slideshow/human-factors-topic-1-introduction/18379460#11
- https://www.slideshare.net/slideshow/human-factors-7950513/7950513
- **Human Factor Printed by L.N.V.M. Society Group of Institutes**

Lecture-5

Purpose of Information Dissemination

- **Increase Awareness**: Sharing important information with stakeholders, employees, or the public to keep them informed.
- **Promote Action**: Encouraging the audience to take specific actions based on the information shared.
- **Facilitate Decision-Making**: Providing key data to help individuals and organizations make informed decisions.
- **Compliance and Reporting**: Disseminating information for regulatory purposes, including reports, updates, and policy changes.

Types of Information to Be Disseminated

- **Research Findings**: Academic papers, studies, and technical reports shared within a professional community.
- **Project Updates**: Progress reports and status updates shared with stakeholders or team members.
- **Policy Changes**: Updates on new regulations, laws, or company policies.
- **News and Announcements**: Updates on organizational changes, industry news, or public notices.
- **Training and Educational Content**: Workshops, manuals, and resources designed to inform or train employees or the public.

Methods of Disseminating Information

a) Traditional Methods

- **Printed Materials**: Brochures, posters, newsletters, and manuals handed out physically.
- **Face-to-Face Communication**: Meetings, seminars, or conferences where information is communicated directly.
- **Telephone**: Direct communication through calls to provide or confirm information.

b) Digital Methods

- **Emails & Newsletters**: Common for formal communication, updates, and newsletters.

Notes By- Karishma Mittal (AP, Civil Engg Department, MITRC, Alwar)

- **Websites and Blogs**: Frequently used to provide in-depth information that can be accessed at any time.
- **Social Media**: Platforms like Twitter, Facebook, LinkedIn to reach large audiences quickly.
- **Webinars & Online Workshops**: Virtual events to disseminate information interactively.
- **Cloud-Based Platforms**: Collaborative tools like Google Drive or SharePoint where documents, updates, and resources are shared.

c) Interactive Methods

- **Surveys and Feedback Forms**: To gauge the effectiveness of information dissemination and gather insights.
- **Discussion Forums & Online Communities**: Engaging stakeholders in discussions via platforms like Slack, Reddit, or industry-specific forums.
- **Public Presentations**: Conferences or public forums where information is presented to an audience with opportunities for feedback.

Challenges in Information Dissemination

- **Information Overload**: The risk of overwhelming the audience with too much information, which may cause confusion or disengagement.
- **Lack of Engagement**: Audience members may not engage with the information if it's not presented in an appealing or accessible format.
- **Cultural or Language Barriers**: Diverse audiences may require different methods of communication, especially when languages or cultural contexts vary.
- **Technical Barriers**: Not everyone may have access to digital tools or technology, especially in remote or underserved areas.
- **Misinterpretation**: Information may be misinterpreted due to ambiguous language, lack of context, or inadequate details

Reference

- https://www.slideshare.net/slideshow/human-factors-topic-1-introduction/18379460#11
- https://www.slideshare.net/slideshow/human-factors-7950513/7950513
- **Human Factor Printed by L.N.V.M. Society Group of Institutess**